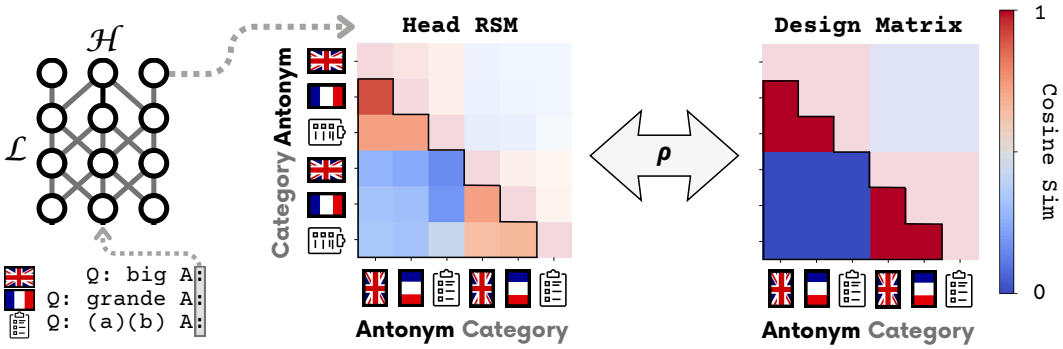

Figure 2: *Representational Similarity Analysis (RSA)*. For each attention head, we compute a representational similarity matrix (RSM) over prompts spanning concepts and input formats (cosine similarity of head outputs). We construct a binary design matrix that marks pairs sharing the same concept, independent of format. The RSA score for a head is Spearman's ρ between the lower-triangular entries of the RSM and the design matrix; higher ρ indicates stronger concept-invariant encoding.

2.1.4 Representational Similarity Analysis

To find attention heads encoding concepts invariant to input formats, we employ representational similarity analysis (RSA; Kriegeskorte (2008)).

For each attention head $a_{\ell j}$ we compute representational similarity matrices (RSMs) where v_i denotes the output extracted from $a_{\ell j}$ for the ith prompt $p_i \in P_N$, and $\theta(\cdot, \cdot)$ is a cosine similarity function.

$$\text{RSM} = \begin{bmatrix} 1 & \cdots & \theta(v_1, v_N) \\ \vdots & \ddots & \vdots \\ \theta(v_N, v_1) & \cdots & 1 \end{bmatrix} \tag{3}$$

We then construct a binary design matrix, DM, where each entry is set to 1 if the corresponding pair of prompts share the same attribute value, and 0 otherwise. In this paper, we consider two attributes: (1) `concept` - does a pair of prompts illustrate the same concept, regardless of the input format? and (2) `prompt_format` - does a pair of prompts have the same question type (i.e. open-ended or multiple-choice)?

We then quantify the alignment between the RSM and DM for the lower-triangles (since similarity matrices are symmetric) using the non-parametric Spearman's rank correlation coefficient ($\boldsymbol{\rho}$).

To localize attention heads carrying invariant concept information we compute the RSA for each attention head obtaining a single Concept RSA score for each attention head.

$$\text{Concept-RSA}(a_{\ell j}) = \rho(\text{RSM}_{\ell j}, \text{Concept-DM}) \tag{4}$$

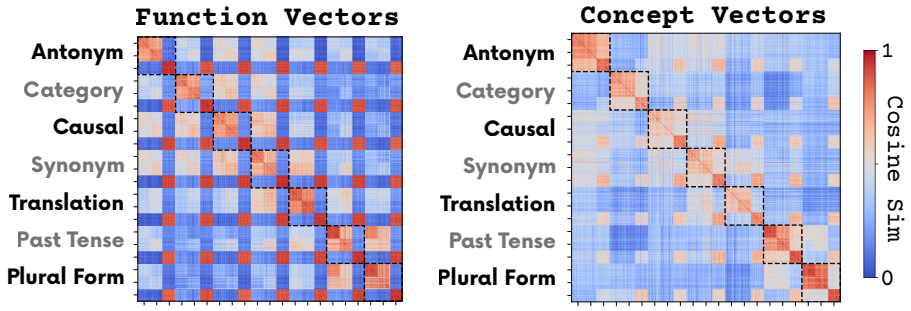

Figure 3: *Similarity matrices*. Full similarity matrices extracted from top $K = 5$ heads in $\mathcal{CV}$s and $\mathcal{FV}$s in Llama 3.1 70B for all concepts. See Appendix B for other models.

2.1.5 Function & Concept Vectors

To form Function/Concept Vectors we create sets of top K ranking attention heads, $\mathcal{A}_{\mathcal{FV}}$ and $\mathcal{A}_{\mathcal{CV}}$, based on their AIE and RSA scores respectively. Function/Concept Vectors for prompt i are then computed as the sum of activations for this prompt, $\mathbf{a}^i_{\ell j}$, from the sets $\mathcal{A}_{\mathcal{FV}}$ and $\mathcal{A}_{\mathcal{CV}}$ respectively.

$$\mathcal{FV}_i = \sum_{a^i_{\ell j} \in \mathcal{A}_{\mathcal{FV}}} \mathbf{a}^i_{\ell j} \quad \mathcal{CV}_i = \sum_{a^i_{\ell j} \in \mathcal{A}_{\mathcal{CV}}} \mathbf{a}^i_{\ell j} \tag{5}$$

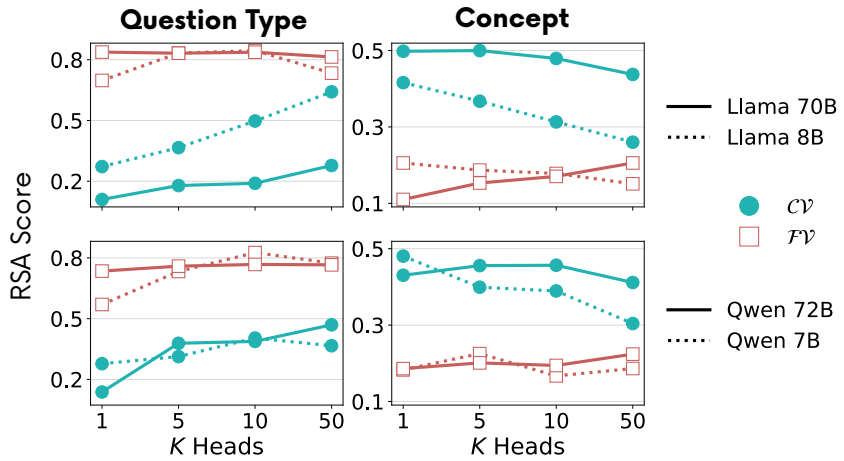

Figure 4: *Concept vs. format RSA*. Question type and Concept RSA scores for $\mathcal{CV}$s and $\mathcal{FV}$s in all models. **Takeaway**: $\mathcal{CV}$s encode more concept information and less input format than $\mathcal{FV}$s.

2.2 Results

2.2.1 Concept Vectors are More Invariant to Input Format

We test invariance to input format by computing RSA with design matrices for concept and question type (following the setup in §2.1.4). We form $\mathcal{FV}$s/$\mathcal{CV}$s by summing the top-K heads ranked by AIE/RSA (Eq. 5). Across models and K, $\mathcal{CV}$s show higher concept RSA and lower question-type RSA than $\mathcal{FV}$s (Figure 4), indicating that $\mathcal{FV}$s encode format more strongly while $\mathcal{CV}$s track concept. Consistently, similarity matrices for Llama 3.1 70B cluster by concept across formats for $\mathcal{CV}$s, but by format for $\mathcal{FV}$s (Figure 3), where within-format type $\mathcal{FV}$ clusters are nearly identical with mean cosine similarity $= 0.90$. $\mathcal{CV}$s nonetheless exhibit a weaker within-format type cluster (mean cosine similarity $= 0.55$), suggesting they retain some low-level format information. Overall, however, $\mathcal{CV}$s remain markedly more invariant to input format than $\mathcal{FV}$s.

Model	K=3	K=5	K=10	K=20	K=50	K=100
Llama-3.1 8B	0	0	1	1	**12**	**28**
Llama-3.1 70B	0	0	0	0	1	**6**
Qwen2.5 7B	0	0	0	**4**	**15**	**39**
Qwen2.5 72B	0	0	0	1	**3**	**13**

Table 1: *RSA–AIE head overlap*. Overlap between RSA and AIE heads (number of overlapping heads among top-K). Bold numbers indicate overlap significantly above chance ($p < 0.05$; details in Appendix E). **Takeaway**: $\mathcal{FV}$s and $\mathcal{CV}$s are composed of different attention heads.

2.2.2 Function & Concept Vectors are Composed of Different Attention Heads

If we compare which heads are selected by the two procedures, we see that $\mathcal{FV}$s and $\mathcal{CV}$s are composed of different attention heads. First, we ranked each head for each method, i.e., AIE (§2.1.3)

for $\mathcal{FV}$s and by Concept-RSA (§2.1.4) for $\mathcal{CV}$s. Then we examined depth and top-K overlap. Layer-averaged scores show similar layer profiles (Figure 5), but head identities barely overlap: for $K \leq 20$ the intersection is near zero and stays small at larger K (Table 1). We also note that AIE scores are highly sparse: their histogram peaks at zero with a long right tail (Figure 12)—so only a few heads have measurable causal effect. Together this supports that AIE-selected *causal* heads are largely distinct from the *invariant*, RSA-selected heads.

To ensure this separation is not an artifact of patching within the same format, we also performed cross-format activation patching (e.g., extracting activations from open-ended prompts and patching them into multiple-choice). This procedure continued to identify the same FV heads and did not identify CV heads (see Appendix O), confirming that FVs are the primary causal drivers regardless of input format.

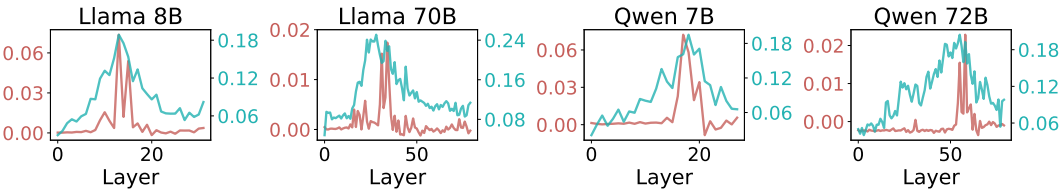

Figure 5: *Layer-wise AIE vs. RSA*. AIE and RSA scores averaged across all heads per layer. **Takeaway**: $\mathcal{FV}$ and $\mathcal{CV}$ heads are in similar layers.

3 Can Concept Vectors Steer?

We now test whether these invariant heads can steer: we introduce how we construct vectors, the AmbiguousICL setup with conflicting cues, and the intervention protocol. $\mathcal{FV}$s win in-distribution; $\mathcal{CV}$s transfer better out-of-distribution with fewer format artifacts, at a cost of smaller gains.

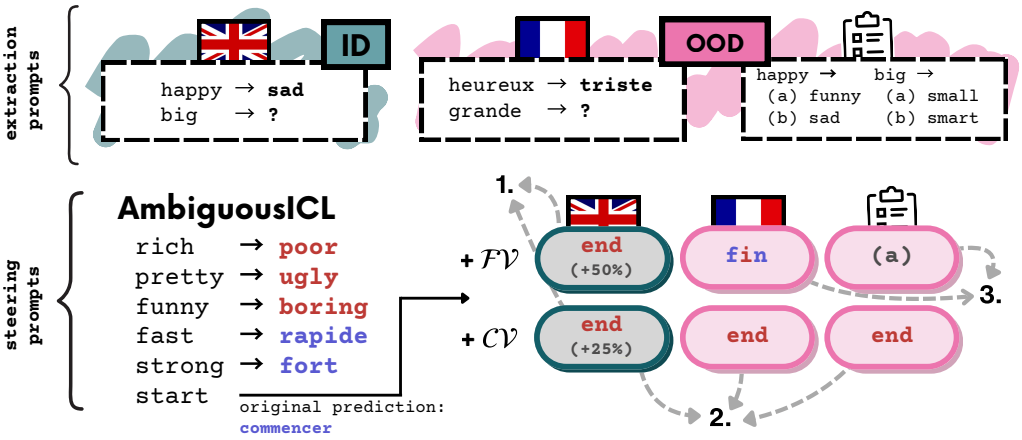

Figure 6: *Overview of steering results*. **Top**: We extract $\mathcal{CV}$s and $\mathcal{FV}$s from antonym ICL prompts in formats that are in-distribution (ID; `OE-ENG`) or out-of-distribution (OOD; `OE-FR`, `MC`) relative to the AmbiguousICL task (bottom-left). **Bottom-left**: We interleave two concepts—**antonym** and **EN→FR translation**—within one prompt; the model's original prediction is the French translation. **Bottom-right**: Predictions after steering. **Takeaways**: (1) $\mathcal{FV}$s yield larger ID gains. (2) $\mathcal{CV}$s show more stable OOD effects across formats. (3) $\mathcal{FV}$s can conflate concept with input format (e.g., French version of antonym and multiple-choice formatting).

3.1 Steering Methods

Steering Vectors Construction. For each concept and input format (`OE-ENG`, `OE-FR`, `MC`), we compute for every selected head $a_{\ell j}$ the mean last-token activation across the 50 *extraction prompts*